VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Locating Log Files

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Objectives

- Get familiar with the structure and content of Linux log files.
- Locate log files in an unfamiliar Linux system.
- Monitor log files in real time using tail and journalctl.

Part 1 – Log File Overview

Step 1a – Apache Web Server Log Entry Analysis

The following single log entry was examined:

```
[Wed Mar 22 11:23:12.207022 2017] [core:error] [pid 3548:tid 4682351596]
[client 209.165.200.230] File does not exist:
/var/www/apache/htdocs/favicon.ico
```

Field	Value	Meaning
Timestamp	Wed Mar 22 11:23:12.207022 2017	Date and time the event occurred
Type	core:error	An error-level event from the core Apache module
PID / TID	3548 / 4682351596	Process ID and Thread ID handling the request
Client IP	209.165.200.230	IP address of the requesting client
Description	File does not exist: /var/www/apache/htdocs/favicon.ico	The browser requested favicon.ico but the file was not found on the server

Answer – What happened?

On Wednesday, March 22, 2017 at 11:23:12, the Apache web server received a request from client 209.165.200.230 for the file /var/www/apache/htdocs/favicon.ico. Apache could not find the file on disk and logged a core:error event. This is a common, non-critical error that occurs when a browser automatically requests a site's favicon and the file has not been placed in the web root directory.

Step 1b – cat /var/log/logstash-tutorial.log

```
[analyst@secOps ~]$ cat /var/log/logstash-tutorial.log
```

```
81.220.24.207 - - [04/Jan/2015:05:24:57 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11 (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
81.220.24.207 - - [04/Jan/2015:05:24:57 +0000] "GET /images/web/2009/banner.png HTTP/1.1" 200 52315 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11 (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
81.220.24.207 - - [04/Jan/2015:05:24:58 +0000] "GET /favicon.ico HTTP/1.1" 200 3638 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11 (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
66.249.73.135 - - [04/Jan/2015:05:25:05 +0000] "GET /blog/geekery/vmware-cpu-performance.html HTTP/1.1" 200 12908 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile/10A5376e Safari/8536.25 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)"
46.105.14.53 - - [04/Jan/2015:05:26:17 +0000] "GET /blog/tags/puppet?flav=rss20 HTTP/1.1" 200 14872 "-" "UniversalFeedParser/4.2-pre-314-svn +http://feedparser.org/"
218.30.103.62 - - [04/Jan/2015:05:27:05 +0000] "GET /robots.txt HTTP/1.1" 200 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:27:10 +0000] "GET /robots.txt HTTP/1.1" 200 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:27:15 +0000] "GET /projects/fex/ HTTP/1.1" 200 14352 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
74.125.40.20 - - [04/Jan/2015:05:27:22 +0000] "GET /?flav=rss20 HTTP/1.1" 200 29941 "-" "FeedBurner/1.0 (http://www.FeedBurner.com)"
71.212.224.97 - - [04/Jan/2015:05:27:34 +0000] "GET /projects/xdotool/ HTTP/1.1" 200 12292 "http://suckless.org/rocks" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:34 +0000] "GET /reset.css HTTP/1.1" 200 1015 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /images/jordan-80.png HTTP/1.1" 200 6146 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /images/web/2009/banner.png HTTP/1.1" 200 52315 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
218.30.103.62 - - [04/Jan/2015:05:27:36 +0000] "GET /projects/xdotool/xdotool.xhtml HTTP/1.1" 304 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
108.174.55.234 - - [04/Jan/2015:05:27:45 +0000] "GET /?flav=rss20 HTTP/1.1" 200 29941 "-" "-"
218.30.103.62 - - [04/Jan/2015:05:27:57 +0000] "GET /blog/geekery/c-vs-python-bdb.html HTTP/1.1" 200 11388 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
121.107.188.202 - - [04/Jan/2015:05:27:57 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171717 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
218.30.103.62 - - [04/Jan/2015:05:28:21 +0000] "GET /blog/productivity/better-zsh-xterm-title-fix.html HTTP/1.1" 200 10185 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:28:43 +0000] "GET /blog/geekery/xvfb-firefox.html HTTP/1.1" 200 10975 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:29:06 +0000] "GET /blog/geekery/puppet-facts-into-mcollective.html HTTP/1.1" 200 9872 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
198.46.149.143 - - [04/Jan/2015:05:29:13 +0000] "GET /blog/geekery/disabling-battery-in-ubuntu-vms.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+semicomplete%2Fmain+%28semicomplete.com+-+Jordan+Sissel%29 HTTP/1.1" 200 9316 "-" "Tiny Tiny RSS/1.11 (http://tt-rss.org/)"
198.46.149.143 - - [04/Jan/2015:05:29:13 +0000] "GET /blog/geekery/solving-good-or-bad-problems.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+semicomplete%2Fmain+%28semicomplete.com+-+Jordan+Sissel%29 HTTP/1.1" 200 10756 "-" "Tiny Tiny RSS/1.11 (http://tt-rss.org/)"
218.30.103.62 - - [04/Jan/2015:05:29:26 +0000] "GET /blog/geekery/jquery-interface-puffer.html%20target= HTTP/1.1" 200 202 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:29:48 +0000] "GET /blog/geekery/ec2-reserved-vs-ondemand.html HTTP/1.1" 200 11834 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
66.249.73.135 - - [04/Jan/2015:05:30:06 +0000] "GET /blog/web/firefox-scrolling-fix.html HTTP/1.1" 200 8956 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile/10A5376e Safari/8536.25 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /projects/xdotool/ HTTP/1.1" 200 12292 "http://www.haskell.org/haskellwiki/Xmonad/Frequently_asked_questions" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /reset.css HTTP/1.1" 200 1015 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
[analyst@secOps ~]$
```

Answer – Is this still a web transaction? Why is the format different?

Yes, the entries in logstash-tutorial.log still represent web transactions — each line records an HTTP request made by a client, including the client IP address, timestamp, HTTP method, requested resource, response code, bytes transferred, referrer URL, and user-agent string.

The format appears different from the Apache error-log entry in item (a) because this file uses the Combined Log Format (a common access-log format), whereas the entry in (a) is from Apache's error log, which uses a distinct format to capture error-specific fields such as severity level, process/thread IDs, and a human-readable description. Access logs and error logs serve different purposes and therefore use different formats.

Step 2a – sudo more /var/log/messages

```
[analyst@secOps ~]$ sudo more /var/log/messages
```

```
Apr 12 10:37:18 labvm systemd[1]: logrotate.service: Deactivated s
uccessfully.
Apr 12 10:37:18 labvm systemd[1]: Finished Rotate log files.
Apr 12 10:37:18 labvm systemd[1]: logrotate.service: Consumed 1.21
1s CPU time.
Apr 12 10:37:20 labvm systemd[1]: man-db.service: Deactivated succ
essfully.
Apr 12 10:37:20 labvm systemd[1]: Finished Daily man-db regenerati
on.
Apr 12 10:37:20 labvm systemd[1]: man-db.service: Consumed 1.770s
CPU time.
Apr 12 10:37:23 labvm kernel: [92257.770551] 10:37:23.868605 times
ync vgsvcTimeSyncWorker: Radical guest time change: 76 116 025 301
 000ns (GuestNow=1 775 990 243 868 581 000 ns GuestLast=1 775 914
127 843 280 000 ns fSetTimeLastLoop=true)
Apr 12 10:41:06 labvm ntpd[648]: 160.119.193.252 local addr 10.0.2
.15 -> <null>
--More--(6%)
```

Answer – Evidence of slow network around 4:20 am on May 19?

Based on the log entries provided in the lab, there is no direct evidence of a slow network event at 4:20 am on May 19. The sample output shown covers dates in March, not May 19.

However, the entries from Mar 20 14:28:29 – 14:29:05 demonstrate a pattern of repeated link-down / link-up cycling on interface enp0s3 (the network adapter), which would cause intermittent connectivity and appear as slowness to users. If similar link-flapping entries existed around 4:20 am on May 19, those would constitute evidence of the reported network issue. To investigate, one would search the log for entries timestamped 'May 19' between 04:15 and 04:25 containing 'link down' or 'link up'.

Part 2 – Locating Log Files in Unknown Systems

Step a – man nginx

```
[analyst@secOps ~]$ man nginx
```

```
NGINX(8)              BSD System Manager's Manual          NGINX(8)

NAME
     nginx — HTTP and reverse proxy server, mail proxy server

SYNOPSIS
     nginx [-?hqTtVv] [-c file] [-g directives] [-p prefix]
           [-s signal]

DESCRIPTION
     nginx (pronounced “engine x”) is an HTTP and reverse proxy
     server, a mail proxy server, and a generic TCP/UDP proxy
     server.  It is known for its high performance, stability,
     rich feature set, simple configuration, and low resource
     consumption.

     The options are as follows:

 Manual page nginx(8) line 1 (press h for help or q to quit)
```

```
The options are as follows:

-?, -h         Print help.

-c file        Use an alternative configuration file.

-g directives  Set global configuration directives.  See
               EXAMPLES for details.

-p prefix      Set the prefix path.  The default value is
               /usr/share/nginx.

-q             Suppress non-error messages during configu-
               ration testing.

-s signal      Send a signal to the master process.  The
               argument signal can be one of: stop, quit,
               reopen, reload.  The following table shows
               the corresponding system signals:

               stop    SIGTERM
               quit    SIGQUIT
               reopen  SIGUSR1
               reload  SIGHUP

-T             Same as -t, but additionally dump configura-
               tion files to standard output.

-t             Do not run, just test the configuration
               file.  nginx checks the configuration file
               syntax and then tries to open files refer-
               enced in the configuration file.

-V             Print the nginx version, compiler version,
               and configure script parameters.

-v             Print the nginx version.
```

```
SIGNALS
     The master process of nginx can handle the following sig-
     nals:

     SIGINT, SIGTERM  Shut down quickly.
     SIGHUP           Reload configuration, start the new worker
                      process with a new configuration, and
                      gracefully shut down old worker processes.
     SIGQUIT          Shut down gracefully.
     SIGUSR1          Reopen log files.
     SIGUSR2          Upgrade the nginx executable on the fly.
     SIGWINCH         Shut down worker processes gracefully.

     While there is no need to explicitly control worker pro-
     cesses normally, they support some signals too:

     SIGTERM          Shut down quickly.
     SIGQUIT          Shut down gracefully.
     SIGUSR1          Reopen log files.

DEBUGGING LOG
     To enable a debugging log, reconfigure nginx to build with
     debugging:

           ./configure --with-debug ...

     and then set the debug level of the error_log:

           error_log /path/to/log debug;

     It is also possible to enable the debugging for a particu-
     lar IP address:

           events {
                   debug_connection 127.0.0.1;
           }

ENVIRONMENT
     The NGINX environment variable is used internally by nginx
     and should not be set directly by the user.

FILES
     /run/nginx.pid
             Contains the process ID of nginx.  The contents of
             this file are not sensitive, so it can be world-
```

```
ENVIRONMENT
     The NGINX environment variable is used internally by nginx
     and should not be set directly by the user.

FILES
     /run/nginx.pid
             Contains the process ID of nginx.  The contents of
             this file are not sensitive, so it can be world-
             readable.

     /etc/nginx/nginx.conf
             The main configuration file.

     /var/log/nginx/error.log
             Error log file.

EXIT STATUS
     Exit status is 0 on success, or 1 if the command fails.

EXAMPLES
     Test configuration file ~/mynginx.conf with global direc-
     tives for PID and quantity of worker processes:

           nginx -t -c ~/mynginx.conf \
                   -g "pid /var/run/mynginx.pid; worker_processes 2;"

SEE ALSO
     Documentation at http://nginx.org/en/docs/.

     For questions and technical support, please refer to
     http://nginx.org/en/support.html.

HISTORY
     Development of nginx started in 2002, with the first public
     release on October 4, 2004.

AUTHORS
     Igor Sysoev <igor@sysoev.ru>.

     This manual page was originally written by Sergey A. Osokin
     <osa@FreeBSD.org.ru> as a result of compiling many nginx
     documents from all over the world.

BSD                        December 5, 2019                      BSD
 Manual page nginx(8) line 91/135 (END) (press h for help or q to quit)
```

Step d – ps ax | grep nginx

```
[analyst@secOps ~]$ ps ax | grep nginx
```

```
[analyst@secOps ~]$ ps ax | grep nginx
  2283 ?        Ss     0:00 nginx: master process /usr/sbin/nginx
  2284 ?        S      0:00 nginx: worker process
  3174 pts/2    S+     0:00 grep nginx
```

Step e – ls /etc/

```
[analyst@secOps ~]$ ls /etc/
```

```
[analyst@secOps ~]$ ls /etc/
adduser.conf                      mime.types
alsa                              mke2fs.conf
alternatives                      modprobe.d
anacrontab                        modules
apache2                           modules-load.d
apm                               mtab
apparmor                          multipath
apparmor.d                        nanorc
apport                            netconfig
apt                               netplan
bash.bashrc                       network
bash_completion                   networkd-dispatcher
bash_completion.d                 NetworkManager
bindresvport.blacklist            networks
binfmt.d                          newt
byobu                             nftables.conf
ca-certificates                   nginx
ca-certificates.conf              nsswitch.conf
ca-certificates.conf.dpkg-old     ntp.conf
cloud                             oinkmaster.conf
compizconfig                      openvswitch
console-setup                     opt
cron.d                            os-release
cron.daily                        PackageKit
cron.hourly                       pam.conf
cron.monthly                      pam.d
crontab                           papersize
cron.weekly                       passwd
crypttab                          passwd-
dbus-1                            perl
dconf                             pm
debconf.conf                      pnm2ppa.conf
debian_version                    polkit-1
default                           pollinate
deluser.conf                      ppp
depmod.d                          profile
dhcp                              profile.d
dictionaries-common               protocols
dkms                              pulse
dpkg                              python3
e2scrub.conf                      python3.10
emacs                             rc0.d
environment                       rc1.d
environment.d                     rc2.d
ethertypes                        rc3.d
```

```
firefox                        rc4.d
fonts                          rc5.d
freeradius                     rc6.d
freetds                        rc.local
fstab                          rcS.d
ftpusers                       resolv.conf
fuse.conf                      rmt
gai.conf                       rpc
ghostscript                    rsyslog.conf
gimp                           rsyslog.d
glvnd                          scite
gnome                          screenrc
groff                          security
group                          selinux
group-                         sensors3.conf
grub.d                         sensors.d
gshadow                        services
gshadow-                       shadow
gss                            shadow-
gtk-2.0                        shells
gtk-3.0                        skel
guest-session                  smi.conf
hdparm.conf                    snort
host.conf                      sos
hostname                       sound
hosts                          speech-dispatcher
hosts.allow                    ssh
hosts.deny                     ssl
ifplugd                        subgid
inetd.conf                     subgid-
init                           subuid
init.d                         subuid-
initramfs-tools                sudo.conf
inputrc                        sudoers
iproute2                       sudoers.d
issue                          sudo_logsrvd.conf
issue.net                      sysctl.conf
java-11-openjdk                sysctl.d
john                           systemd
kernel                         terminfo
landscape                      timezone
ldap                           tlp.d
ld.so.cache                    tmpfiles.d
ld.so.conf                     ubuntu-advantage
ld.so.conf.d                   ucf.conf
legal                          udev
```

```
libao.conf                    udisks2
libaudit.conf                 ufw
libblockdev                   update-manager
libnl-3                       update-motd.d
libpaper.d                    update-notifier
libreoffice                   UPower
lightdm                       vdpau_wrapper.cfg
lighttpd                      vim
locale.alias                  vmware-tools
locale.gen                    vsftpd.conf
localtime                     vtrgb
logcheck                      wgetrc
login.defs                    wireshark
logrotate.conf                wpa_supplicant
logrotate.d                   X11
lsb-release                   xattr.conf
machine-id                    xdg
magic                         xinetd.conf
magic.mime                    xinetd.d
manpath.config                xrdb
mate-settings-daemon          zsh_command_not_found
```

Step f – ls -l /etc/nginx/

```
[analyst@secOps ~]$ ls -l /etc/nginx/
```

```
[analyst@secOps ~]$ ls -l /etc/nginx/
total 68
drwxr-xr-x 2 root root 4096 Nov 10  2022 conf.d
-rw-r--r-- 1 root root 2812 Feb 10  2023 custom_server.conf
-rw-r--r-- 1 root root 1125 Jul 27  2022 fastcgi.conf
-rw-r--r-- 1 root root 1055 Jul 27  2022 fastcgi_params
-rw-r--r-- 1 root root 2837 Jul 27  2022 koi-utf
-rw-r--r-- 1 root root 2223 Jul 27  2022 koi-win
-rw-r--r-- 1 root root 3957 Aug  2  2022 mime.types
drwxr-xr-x 2 root root 4096 Nov 10  2022 modules-available
drwxr-xr-x 2 root root 4096 Feb 10  2023 modules-enabled
-rw-r--r-- 1 root root 1447 Jul 27  2022 nginx.conf
-rw-r--r-- 1 root root  180 Jul 27  2022 proxy_params
-rw-r--r-- 1 root root  636 Jul 27  2022 scgi_params
drwxr-xr-x 2 root root 4096 Feb 10  2023 sites-available
drwxr-xr-x 2 root root 4096 Feb 10  2023 sites-enabled
drwxr-xr-x 2 root root 4096 Feb 10  2023 snippets
-rw-r--r-- 1 root root  664 Jul 27  2022 uwsgi_params
-rw-r--r-- 1 root root 3071 Jul 27  2022 win-utf
```

Step g – cat /etc/nginx/nginx.conf

```
[analyst@secOps ~]$ cat /etc/nginx/nginx.conf
```

```
[analyst@secOps ~]$ cat /etc/nginx/nginx.conf
user www-data;
worker_processes auto;
pid /run/nginx.pid;
include /etc/nginx/modules-enabled/*.conf;

events {
        worker_connections 768;
        # multi_accept on;
}

http {

        ##
        # Basic Settings
        ##

        sendfile on;
        tcp_nopush on;
        types_hash_max_size 2048;
        # server_tokens off;

        # server_names_hash_bucket_size 64;
        # server_name_in_redirect off;

        include /etc/nginx/mime.types;
        default_type application/octet-stream;

        ##
        # SSL Settings
        ##

        ssl_protocols TLSv1 TLSv1.1 TLSv1.2 TLSv1.3; # Dropping SSLv3, ref: POODLE
        ssl_prefer_server_ciphers on;

        ##
        # Logging Settings
        ##

        access_log /var/log/nginx/access.log;
        error_log /var/log/nginx/error.log;

        ##
        # Gzip Settings
        ##
```

```
        gzip on;

        # gzip_vary on;
        # gzip_proxied any;
        # gzip_comp_level 6;
        # gzip_buffers 16 8k;
        # gzip_http_version 1.1;
        # gzip_types text/plain text/css application/json application/javascript text/xml application/xml application/xml+rss text/javascript;

        ##
        # Virtual Host Configs
        ##

        include /etc/nginx/conf.d/*.conf;
        include /etc/nginx/sites-enabled/*;
}


#mail {
#       # See sample authentication script at:
#       # http://wiki.nginx.org/ImapAuthenticateWithApachePhpScript
#
#       # auth_http localhost/auth.php;
#       # pop3_capabilities "TOP" "USER";
#       # imap_capabilities "IMAP4rev1" "UIDPLUS";
#
#       server {
#               listen     localhost:110;
#               protocol   pop3;
#               proxy      on;
#       }
#
#       server {
#               listen     localhost:143;
#               protocol   imap;
#               proxy      on;
#       }
#}
```

Step h – ls -l /var/log/

```
[analyst@secOps ~]$ ls -l /var/log/
```

```
[analyst@secOps ~]$ ls -l /var/log/
total 2228
-rw-r--r--  1 root             root               0 Jan  2 00:00 alternatives.log
-rw-r--r--  1 root             root           46356 Feb 10  2023 alternatives.log.1
drwxr-xr-x  2 root             root            4096 Jan  2 00:00 apt
-rw-r-----  1 syslog           adm             4450 Apr 12 22:46 auth.log
-rw-r-----  1 syslog           adm             5482 Apr 10 21:58 auth.log.1
-rw-r-----  1 syslog           adm             1169 Apr  2 12:17 auth.log.2.gz
-rw-r-----  1 syslog           adm              733 Jan 12 01:17 auth.log.3.gz
-rw-r-----  1 syslog           adm             2523 Jan  4 00:06 auth.log.4.gz
-rw-r--r--  1 root             root           64549 Aug  9  2022 bootstrap.log
-rw-rw----  1 root             utmp               0 Apr  6 23:07 btmp
-rw-rw----  1 root             utmp               0 Jan  2 00:00 btmp.1
-rw-r--r--  1 syslog           adm            96706 Feb 10  2023 cloud-init.log
-rw-r-----  1 root             adm             4935 Feb 10  2023 cloud-init-output.log
drwxr-xr-x  2 root             root            4096 Aug  3  2022 dist-upgrade
-rw-r-----  1 root             adm            45386 Apr 10 13:51 dmesg
-rw-r-----  1 root             adm            46130 Apr 10 13:43 dmesg.0
-rw-r-----  1 root             adm            14188 Jan 11 23:17 dmesg.1.gz
-rw-r-----  1 root             adm            14212 Dec 19 06:52 dmesg.2.gz
-rw-r-----  1 root             adm            14453 Feb 10  2023 dmesg.3.gz
-rw-r--r--  1 root             root               0 Jan  2 00:00 dpkg.log
-rw-r--r--  1 root             root         1131973 Apr  9  2025 dpkg.log.1
-rw-r--r--  1 root             root           32288 Apr  9  2025 faillog
-rw-r--r--  1 root             root            3838 Feb 10  2023 fontconfig.log
drwxr-xr-x  2 freerad          adm             4096 Dec 22 00:00 freeradius
-rw-r--r--  1 root             root            1300 Apr 10 13:51 gpu-manager.log
drwxr-x---  3 root             adm             4096 Feb 10  2023 installer
drwxr-sr-x+ 3 root             systemd-journal 4096 Feb 10  2023 journal
-rw-r-----  1 syslog           adm             3882 Apr 12 22:46 kern.log
-rw-r-----  1 syslog           adm           110357 Apr 12 10:37 kern.log.1
-rw-r-----  1 syslog           adm             1905 Apr  6 23:07 kern.log.2.gz
-rw-r-----  1 syslog           adm            13258 Jan 12 01:17 kern.log.3.gz
-rw-r-----  1 syslog           adm            12609 Jan  4 00:06 kern.log.4.gz
drwxr-xr-x  2 landscape        landscape       4096 Feb 10  2023 landscape
-rw-rw-r--  1 root             utmp          294628 Apr  9  2025 lastlog
drwxr-xr-x  2 root             root            4096 Apr 12 10:37 lightdm
-rw-r--r--  1 root             root           24464 Jan 16  2023 logstash-tutorial.log
drwxr-xr-x  2 root             adm             4096 Dec 22 00:00 nginx
-rw-r--r--  1 root             root               0 Jan 16  2023 nginx-logstash.log
drwxr-xr-x  2 ntp              ntp             4096 Feb 16  2022 ntpstats
drwxr-xr-x  2 root             root            4096 Apr 12 10:37 openvswitch
drwx------  2 root             root            4096 Aug  9  2022 private
drwxr-s---  2 snort            adm             4096 Feb 10  2023 snort
drwx------  2 speech-dispatcher root           4096 May 23  2023 speech-dispatcher
-rw-r-----  1 syslog           adm            18462 Apr 12 22:46 syslog
```

```
-rw-r-----  1 syslog   adm      18462 Apr 12 22:46 syslog
-rw-r-----  1 syslog   adm     277607 Apr 12 10:37 syslog.1
-rw-r-----  1 syslog   adm      10749 Apr  6 23:07 syslog.2.gz
-rw-r-----  1 syslog   adm      22364 Jan 12 01:17 syslog.3.gz
-rw-r-----  1 syslog   adm      26470 Jan  4 00:06 syslog.4.gz
-rw-r--r--  1 root     root         0 Jan  2 00:00 ubuntu-advantage.log
-rw-r--r--  1 root     root      1583 Feb 10  2023 ubuntu-advantage.log.1
-rw-r--r--  1 root     root      1150 Apr 12 17:06 ubuntu-advantage-timer.log
-rw-r--r--  1 root     root      2990 Jan 11 23:37 ubuntu-advantage-timer.log.1
-rw-r--r--  1 root     root      1041 Jan  1 23:37 ubuntu-advantage-timer.log.2.gz
drwxr-x---  2 root     adm       4096 Feb 10  2023 unattended-upgrades
-rw-------  1 root     root       678 Feb 10  2023 vboxadd-install.log
-rw-r--r--  1 root     root       157 Feb 10  2023 vboxadd-setup.log
-rw-r-----  1 root     adm          0 Apr  6 23:07 vsftpd.log
-rw-------  1 root     root      1120 Apr  2 12:00 vsftpd.log.1
-rw-rw-r--  1 root     utmp      9216 Apr 10 12:35 wtmp
-rw-r--r--  1 root     root     36977 Apr 12 23:03 Xorg.0.log
-rw-r--r--  1 root     root     23681 Apr 10 11:43 Xorg.0.log.old
```

Step i – sudo ls -l /var/log/nginx

```
[analyst@secOps ~]$ sudo ls -l /var/log/nginx
```

```
[analyst@secOps ~]$ ls -l /var/log/nginx
total 4
-rw-r----- 1 www-data adm  0 Feb 10  2023 access.log
-rw-r----- 1 www-data adm  0 Dec 22 00:00 error.log
-rw-r----- 1 www-data adm 78 Feb 10  2023 error.log.1
```

Part 3 – Monitoring Log Files in Real Time

Step 1a – sudo tail /var/log/nginx/access.log

```
[analyst@secOps ~]$ sudo tail /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET / HTTP/1.1" 200 147 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100
101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100
101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

Step 1b – sudo tail -n 5 /var/log/nginx/access.log

```
[analyst@secOps ~]$ sudo tail -n 5 /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail -n 5 /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100
101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

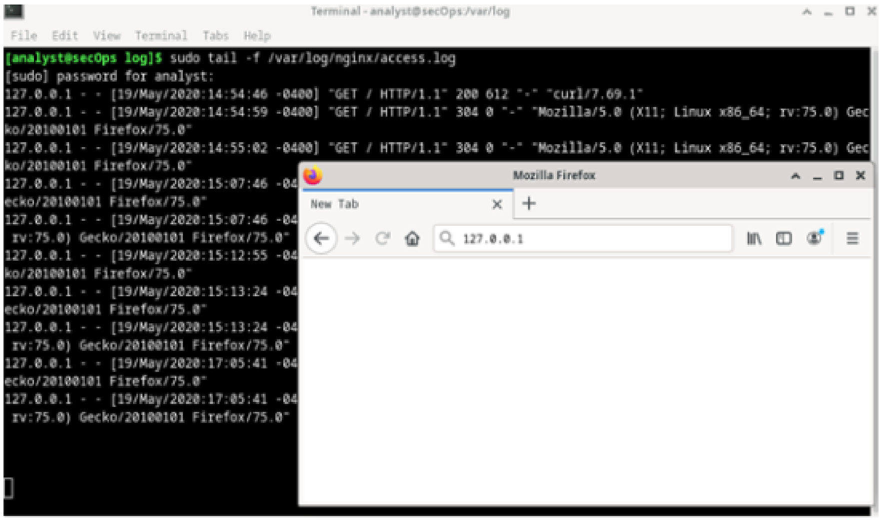
Step 1c – sudo tail -f (Real-Time Monitor)

```
[analyst@secOps ~]$ sudo tail -f /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET / HTTP/1.1" 200 147 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100
101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100
101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0)
Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

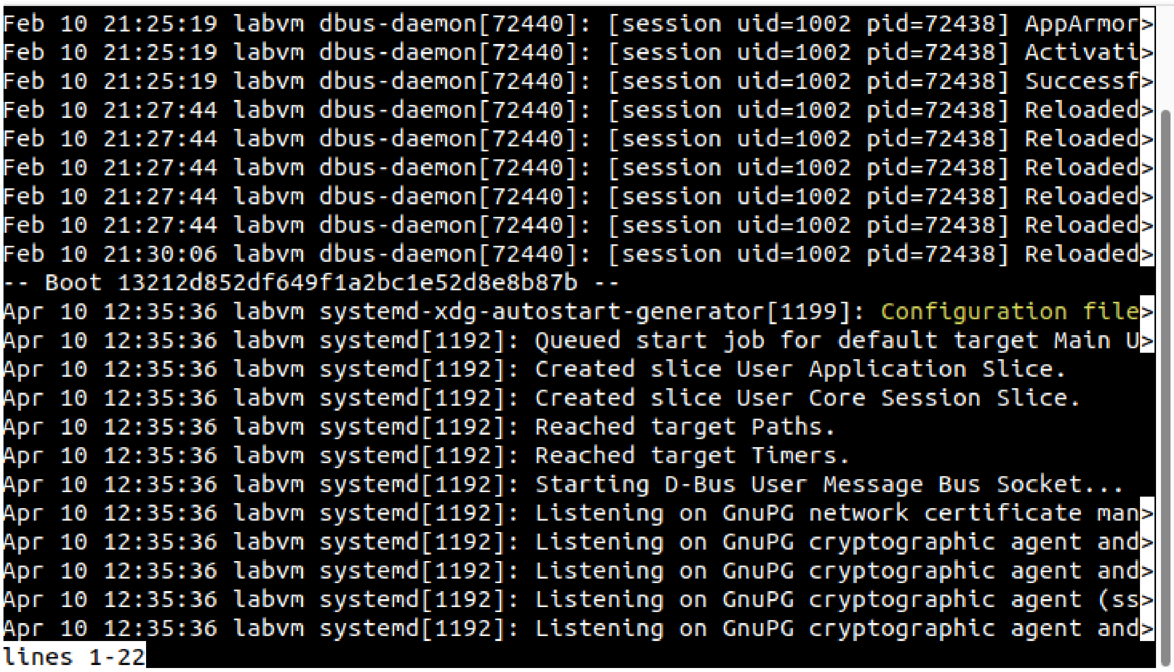
Step 1d & e – Browsing to 127.0.0.1 while tail -f is running

A web browser was opened and navigated to 127.0.0.1. The nginx server served the request and new entries appeared in real time at the bottom of the terminal window running tail -f.



Step 2a – journalctl (no options)

```
[analyst@secOps ~]$ journalctl
```



```
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] AppArmor>
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Activati>
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Successf>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:30:06 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
-- Boot 13212d852df649f1a2bc1e52d8e8b87b --
Apr 10 12:35:36 labvm systemd-xdg-autostart-generator[1199]: Configuration file>
Apr 10 12:35:36 labvm systemd[1192]: Queued start job for default target Main U>
Apr 10 12:35:36 labvm systemd[1192]: Created slice User Application Slice.
Apr 10 12:35:36 labvm systemd[1192]: Created slice User Core Session Slice.
Apr 10 12:35:36 labvm systemd[1192]: Reached target Paths.
Apr 10 12:35:36 labvm systemd[1192]: Reached target Timers.
Apr 10 12:35:36 labvm systemd[1192]: Starting D-Bus User Message Bus Socket...
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG network certificate man>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent (ss>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
lines 1-22
```

Answer – How to run journalctl and see ALL log entries?

Run journalctl as root using sudo:

```
sudo journalctl
```

This grants access to all journal entries, including those belonging to other users and the system. Alternatively, the analyst user could be added to the systemd-journal, adm, or wheel group, which would allow journalctl to display all messages without sudo.

Step 2b – sudo journalctl -b (boot entries)

```
[analyst@secOps ~]$ sudo journalctl -b
```

```
Apr 10 13:51:25 labvm kernel: Linux version 5.15.0-60-generic (buildd@lcy02-amd>
Apr 10 13:51:25 labvm kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-5.15.0-60->
Apr 10 13:51:25 labvm kernel: KERNEL supported cpus:
Apr 10 13:51:25 labvm kernel:   Intel GenuineIntel
Apr 10 13:51:25 labvm kernel:   AMD AuthenticAMD
Apr 10 13:51:25 labvm kernel:   Hygon HygonGenuine
Apr 10 13:51:25 labvm kernel:   Centaur CentaurHauls
Apr 10 13:51:25 labvm kernel:   zhaoxin   Shanghai
Apr 10 13:51:25 labvm kernel: [Firmware Bug]: TSC doesn't count with P0 frequen>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 flo>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: xstate_offset[2]:  576, xstate_sizes[2]:>
Apr 10 13:51:25 labvm kernel: x86/fpu: Enabled xstate features 0x7, context siz>
Apr 10 13:51:25 labvm kernel: signal: max sigframe size: 1776
Apr 10 13:51:25 labvm kernel: BIOS-provided physical RAM map:
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000f>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffe>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fff>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0>
lines 1-22
```

Step 2c – sudo journalctl -b -2 (two boots ago)

```
[analyst@secOps ~]$ sudo journalctl -b -2
```

```
Apr 10 13:51:25 labvm kernel: Linux version 5.15.0-60-generic (buildd@lcy02-amd>
Apr 10 13:51:25 labvm kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-5.15.0-60->
Apr 10 13:51:25 labvm kernel: KERNEL supported cpus:
Apr 10 13:51:25 labvm kernel:   Intel GenuineIntel
Apr 10 13:51:25 labvm kernel:   AMD AuthenticAMD
Apr 10 13:51:25 labvm kernel:   Hygon HygonGenuine
Apr 10 13:51:25 labvm kernel:   Centaur CentaurHauls
Apr 10 13:51:25 labvm kernel:   zhaoxin   Shanghai
Apr 10 13:51:25 labvm kernel: [Firmware Bug]: TSC doesn't count with P0 frequen>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 flo>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: xstate_offset[2]:  576, xstate_sizes[2]:>
Apr 10 13:51:25 labvm kernel: x86/fpu: Enabled xstate features 0x7, context siz>
Apr 10 13:51:25 labvm kernel: signal: max sigframe size: 1776
Apr 10 13:51:25 labvm kernel: BIOS-provided physical RAM map:
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000f>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffe>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fff>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0>
lines 1-22
```

Step 2d – sudo journalctl --list-boots

```
[analyst@secOps ~]$ sudo journalctl --list-boots
```

```
[analyst@secOps ~]$ sudo journalctl --list-boots
-4 cc7dd27d5a8a4df1a4d3457c30f9dd4b Fri 2023-02-10 21:10:50 UTC—Fri 2023-02-10 >
-3 a9b24cc963364eceb493c7c47117432e Fri 2025-12-19 08:52:06 UTC—Wed 2026-01-07 >
-2 fcc2db4f1c554d52a0ea958c6e8b6edc Mon 2026-01-12 01:17:08 UTC—Mon 2026-04-06 >
-1 c285fe5cf8224dc38fd716f36db7a9e0 Fri 2026-04-10 13:43:06 UTC—Fri 2026-04-10 >
 0 13212d852df649f1a2bc1e52d8e8b87b Fri 2026-04-10 13:51:25 UTC—Sun 2026-04-12 >
lines 1-5/5 (END)
```

Step 2e – Filtering by time (--since)

```
[analyst@secOps ~]$ sudo journalctl --since '2 hours ago'
[analyst@secOps ~]$ sudo journalctl --since '1 day ago'
```

```
Apr 12 22:46:27 labvm kernel: 22:46:27.441267 timesync vgsvcTimeSyncWorker: Rad>
Apr 12 22:46:27 labvm CRON[2897]: pam_unix(cron:session): session opened for us>
Apr 12 22:46:27 labvm systemd[1]: Condition check resulted in Run anacron jobs >
Apr 12 22:46:27 labvm CRON[2900]: (root) CMD ([ -x /etc/init.d/anacron ] && if >
Apr 12 22:46:27 labvm CRON[2897]: pam_unix(cron:session): session closed for us>
Apr 12 23:06:34 labvm systemd[1]: Starting Ubuntu Advantage Timer for running r>
Apr 12 23:06:35 labvm systemd[1]: ua-timer.service: Deactivated successfully.
Apr 12 23:06:35 labvm systemd[1]: Finished Ubuntu Advantage Timer for running r>
Apr 12 23:07:34 labvm systemd-resolved[2489]: Clock change detected. Flushing c>
Apr 12 23:07:40 labvm ntpd[648]: Soliciting pool server 196.10.52.58
Apr 12 23:07:41 labvm ntpd[648]: Soliciting pool server 102.205.240.0
Apr 12 23:09:18 labvm sudo[3098]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; US>
Apr 12 23:09:18 labvm sudo[3098]: pam_unix(sudo:session): session opened for us>
Apr 12 23:09:18 labvm sudo[3098]: pam_unix(sudo:session): session closed for us>
Apr 12 23:11:47 labvm sudo[3103]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; US>
Apr 12 23:11:47 labvm sudo[3103]: pam_unix(sudo:session): session opened for us>
Apr 12 23:11:47 labvm sudo[3103]: pam_unix(sudo:session): session closed for us>
Apr 12 23:12:19 labvm sudo[3106]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; US>
Apr 12 23:12:19 labvm sudo[3106]: pam_unix(sudo:session): session opened for us>
Apr 12 23:12:19 labvm sudo[3106]: pam_unix(sudo:session): session closed for us>
Apr 12 23:13:31 labvm wnck-applet[1610]: Negative content width -1 (allocation >
Apr 12 23:13:31 labvm wnck-applet[1610]: Negative content width -1 (allocation >
lines 1-22
```

Step 2f – sudo journalctl -u nginx.service

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service
```

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service
Feb 10 21:29:12 labvm systemd[1]: Starting A high performance web server and a >
Feb 10 21:29:12 labvm systemd[1]: Started A high performance web server and a r>
Feb 10 21:31:17 labvm systemd[1]: Stopping A high performance web server and a >
Feb 10 21:31:17 labvm systemd[1]: nginx.service: Deactivated successfully.
Feb 10 21:31:17 labvm systemd[1]: Stopped A high performance web server and a r>
lines 1-5/5 (END)
```

Step 2g – sudo journalctl -f (real-time follow)

```
[analyst@secOps ~]$ sudo journalctl -f
```

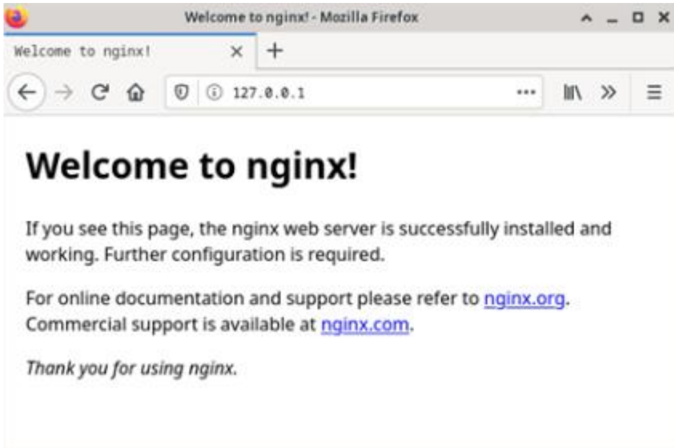
```
[analyst@secOps ~]$ sudo journalctl -f
Apr 12 23:52:54 labvm sudo[3355]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:53:53 labvm sudo[3355]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:54:27 labvm sudo[3359]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl --since 2 hours ago
Apr 12 23:54:27 labvm sudo[3359]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:55:29 labvm sudo[3359]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:56:06 labvm sudo[3363]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl -u nginx.service
Apr 12 23:56:06 labvm sudo[3363]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:57:13 labvm sudo[3363]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:57:28 labvm sudo[3368]:  analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl -f
Apr 12 23:57:28 labvm sudo[3368]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
```

Step 2h & i – journalctl -u nginx.service -f + browser test

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service -f
```

A browser was opened and directed to 127.0.0.1. A missing favicon.ico error appeared in real time in the journalctl output.

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service -f
[sudo] password for analyst:
Feb 10 21:29:12 labvm systemd[1]: Starting A high performance web server and a r
everse proxy server...
Feb 10 21:29:12 labvm systemd[1]: Started A high performance web server and a re
verse proxy server.
Feb 10 21:31:17 labvm systemd[1]: Stopping A high performance web server and a r
everse proxy server...
Feb 10 21:31:17 labvm systemd[1]: nginx.service: Deactivated successfully.
Feb 10 21:31:17 labvm systemd[1]: Stopped A high performance web server and a re
verse proxy server.
```



Reflection & Key Takeaways

- Log files are essential for troubleshooting system and application events. They provide a historical record that allows analysts to trace what happened and when.
- Log file locations follow conventions (e.g., /var/log/), but are ultimately chosen by the developer. Always consult documentation first.
- When documentation is unavailable, a combination of web research and direct system investigation (manual pages, configuration files, directory listings) is effective.
- Correct clock synchronization (NTP) is critical. Inaccurate timestamps make it very difficult to correlate and trace events across multiple systems.
- Events from different sources are often analyzed together. Consistent, accurate timestamps allow log entries from the OS, web server, firewall, and other services to be correlated.
- The tail -f command and journalctl -f are powerful real-time monitoring tools. They allow analysts to observe live system and application behavior without waiting for a static log dump.

Commands Reference Summary

Command	Purpose
`cat /var/log/logstash-tutorial.log`	Display entire log file at once
`sudo more /var/log/messages`	Page through a log file (root required)
`man nginx`	Read nginx manual page for log file hints
`ps ax | grep nginx`	Verify nginx process is running
`ls /etc/  &  ls -l /etc/nginx/`	Locate nginx configuration directory and files
`cat /etc/nginx/nginx.conf`	Read nginx configuration to find log paths
`ls -l /var/log/`	List all log files and directories
`sudo ls -l /var/log/nginx`	List nginx-specific log files (root required)
`sudo tail /var/log/nginx/access.log`	Display last 10 lines of nginx access log
`sudo tail -n 5 /var/log/nginx/access.log`	Display last 5 lines
`sudo tail -f /var/log/nginx/access.log`	Real-time log monitoring (follow mode)
`journalctl`	Display all journald log entries
`sudo journalctl -b`	Display current-boot journal entries
`sudo journalctl -b -2`	Display entries from two boots ago
`sudo journalctl --list-boots`	List all recorded boot sessions

`sudo journalctl --since '2 hours ago'`	Filter entries by time range
`sudo journalctl -u nginx.service`	Show entries for a specific service unit
`sudo journalctl -f`	Real-time journal monitoring
`sudo journalctl -u nginx.service -f`	Real-time monitoring for nginx service